Bala Subramanian

Full Stack Developer

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Professional Summary

Full Stack Developer with experience developing scalable and secure web applications using Next.js, React, Node.js, and the MERN stack. Skilled in REST API development, real-time application architecture, database design, authentication, and role-based access control (RBAC). Experienced in integrating AI-powered features, Large Language Models, Voice AI, and real-time communication systems to build high-performance, user-focused solutions.

Technical Skills

Languages: TypeScript, JavaScript (ES6+), SQL, Python, Java, HTML5, CSS3
Frontend: React.js, Next.js, Redux, Tailwind CSS, Shadcn/UI
Backend: Node.js, Express, Prisma ORM, Convex, Socket.io, RESTful APIs,OAuth2.0, Clerk Auth
Cloud & DB: PostgreSQL, MongoDB, Firebase, Docker, Vercel, AWS, MySQL
Core Competencies: AI Implementation, Real-Time Systems, Web Security, E2E Encryption, RBAC

Experience

Full Stack Developer Intern
Ernestwell (United Kingdom, Remote) **Nov 2024 – Feb 2025**

- Engineered **Unibritend**, a comprehensive study-abroad platform using **React.js (Vite)**, implementing dynamic search and filtering systems for 400+ institutional datasets across the UK, USA, and Canada.
- Developed a secure **Admin Portal** using **Firebase Auth** and **Firestore** to manage college data efficiently.
- Created an interactive **FAQ Chatbot**, reducing manual support queries by 30
- Implemented responsive forms with phone number validation and real-time cloud submission.

Projects

Smart Exam Platform – AI EdTech Platform GitHub
Technologies: Next.js 16, TypeScript, Google Gemini 2.5 Flash, Prisma, PostgreSQL

- Architected a platform utilizing **Gemini 2.5 Flash** for automated high-quality MCQ generation based on specific topics and difficulty levels.
- Developed an AI-driven analytics engine that identifies student weak topics and provides a **Predicted Score** for future attempts based on historical performance and provides suggestions for improvements.
- Implemented a secure **RBAC** system and server-side grading logic using **Prisma ORM** to ensure exam integrity.

SecureChat – Real-Time E2EE Messaging Platform
Technologies: React.js, Node.js, Socket.io, MongoDB GitHub

- Designed a low-latency real-time chat system using **Socket.io** for bidirectional communication.
- Engineered **End-to-End Encryption (E2EE)** using AES-256 on the client side, ensuring zero-knowledge privacy.
- Secured backend APIs with **JWT rotation**, **Bcrypt** hashing, and input sanitization to prevent XSS/NoSQL attacks.

AI Voice Fitness Coach
Technologies: Next.js, Convex, Vapi.ai, Gemini API, TypeScript GitHub

- Built a serverless app integrating **Vapi.ai** for real-time conversational voice interaction and data collection.
- Orchestrated **Convex actions** to process voice data and query **Gemini AI** to generate structured JSON workout plans.
- Enforced full-stack type safety with **TypeScript** and secure user session management via **Clerk**.

AI Career Coach & Resume Analyzer
Technologies: MERN Stack, Groq API, Helmet.js, Google OAuth GitHub

- Integrated **Groq API** with optimized system prompts to deliver instant resume analysis and mock interview questions.
- Implemented **Google OAuth 2.0 (Passport.js)** for frictionless and secure user onboarding.
- Hardened the Express server with **Helmet.js** and **rate-limiting** to mitigate brute-force and DDoS attempts.

Awards & Certifications

- **Patent Filed (2024):** "Animal Sound-Based Audiometry Test Using Software" (IP India) – Achieved 95% diagnostic accuracy without clinical hardware.
- **Certifications:** Network Security, DSA, Cryptography, Red Hat System Administration, and Penetration Testing.

Education

Francis Xavier Engineering College, Tirunelveli **Graduating: 2026**
B.E. Computer Science and Engineering (Minor: Cyber Security) **CGPA: 8.6 / 10.0**